Metals

We have been using metals, either in the form of pure elements or combined with other elements as alloys, for thousands of years to make all sorts of useful objects, from jewelry and cutlery to bridges and spacecraft.

The properties of metals

Metals tend to be strong but malleable, are good conductors of heat and electricity, and have high melting points. However, pure metals tend to be too soft or brittle to be useful. Their properties can often be improved by combining them with other elements to form alloys. Most metals in everyday use are alloys, one of the most common being steel.

Shiny
Metals have many electrons on their surface that absorb and then reemit light, giving metals a shiny appearance.

Good heat conductors
Electrons in metals can move freely, so when they gain heat energy, they can pass it on quickly.

Strong
The atoms in metals are arranged in a regular pattern and strongly bonded together. This makes metals strong.

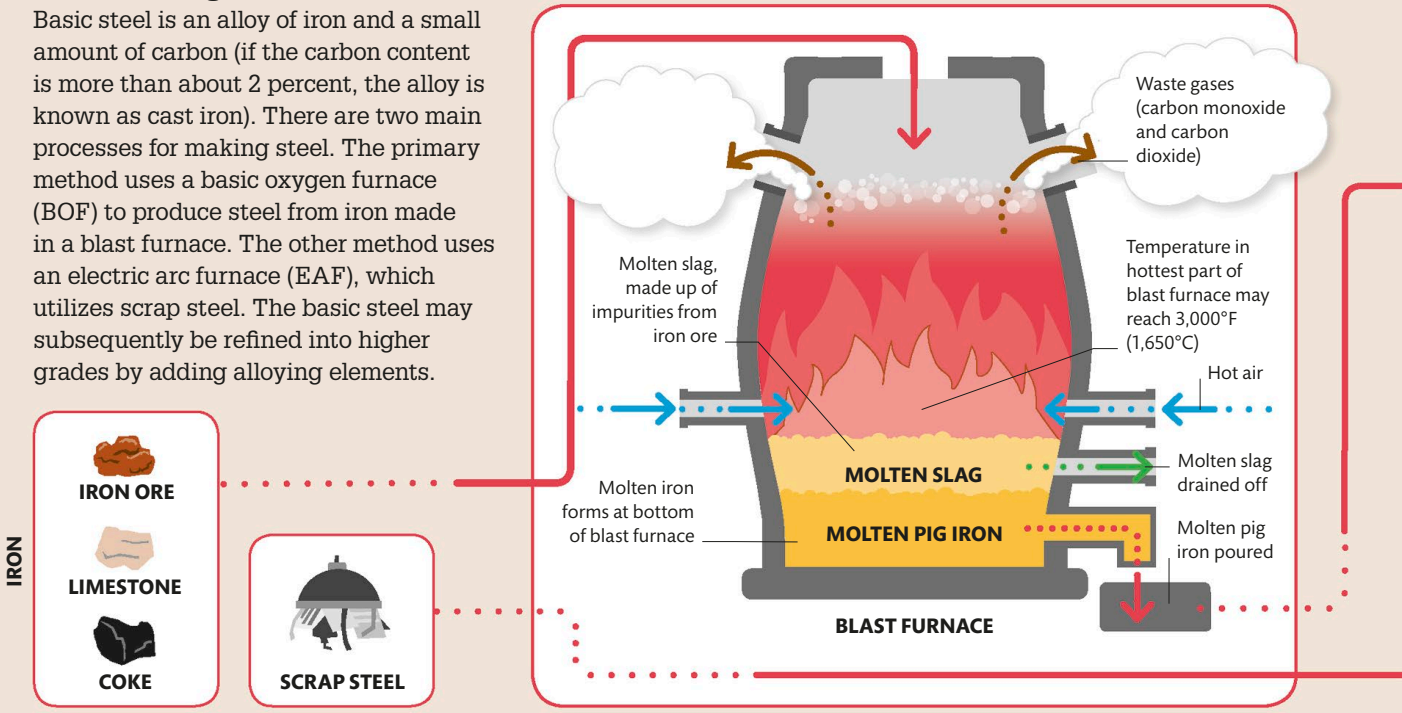
Good electrical conductors
Because electrons in metals can carry electrical charge and move freely, electric currents flow through them easily.

High melting point
The strong bonds between atoms in a metal mean that it takes a lot of heat to free the atoms and melt the metal.

Malleable
The molecular structure of metals allows layers of atoms to slide so that the metal is malleable and can be easily shaped.

Steelmaking

Basic steel is an alloy of iron and a small amount of carbon (if the carbon content is more than about 2 percent, the alloy is known as cast iron). There are two main processes for making steel. The primary method uses a basic oxygen furnace (BOF) to produce steel from iron made in a blast furnace. The other method uses an electric arc furnace (EAF), which utilizes scrap steel. The basic steel may subsequently be refined into higher grades by adding alloying elements.



1 Raw materials
The raw materials for making iron are iron ore (iron oxide plus impurities), limestone (calcium carbonate), and coke (carbon). Steel is produced using iron from a blast furnace, sometimes with scrap steel added, or directly from scrap steel alone.

2 Making iron
In a blast furnace, coke reacts with hot air to produce carbon monoxide, which then reacts with the iron ore to produce pig iron (iron with a high carbon content). The limestone removes most of the impurities from the iron ore. The impurities form a molten slag on top of the molten pig iron.